

Autonomous AI-Agent Marketplace for Micro-SaaS Solopreneurs

Project Workbook

By

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Project summary. The Autonomous AI-Agent Marketplace is a platform on which developers publish purpose-built, autonomous "agentic microservices" (for example: customer-support resolution, dynamic pricing, invoicing and cross-border tax compliance) and on which solopreneurs running micro-SaaS businesses discover, configure, approve and monitor those agents against their own business systems. The project repository is github.com/vineetkumar-sjsu/CMPE_295A_Final_Project_FA26.

Team responsibilities. Each member owns one technical area for the whole project and authored the corresponding workbook sections below.

Member	Primary technical ownership	Workbook sections owned
Nikhil Hiro Ghind	Agent runtime, developer SDK/CLI, reference agents, guardrails & human-in-the-loop, agent evaluation	1.1-1.4 (agents, tools, evaluation), 3.4 (agent requirements), 8.3-8.4
Vineet Kumar	Marketplace back-end services (registry, catalog, orchestration), data layer, cloud infrastructure, CI/CD	1.6-1.7 (platforms, systems), Ch. 5 Architecture, 8.1, Ch. 9 Schedule
Sakshat Nandkumar Patil	Web front-ends (Developer Portal, Solopreneur Console), identity & security, billing, observability dashboards	1.5 (security), State of the Art, Ch. 2 Justification, Ch. 3-4, 8.2

Chapter 1. Literature Search, State of the Art

Literature Search

The project sits at the intersection of four research areas: (i) large-language-model (LLM) based autonomous agents and their ability to use tools and APIs; (ii) the evaluation and security of such agents when they act on real systems; (iii) software marketplaces and two-sided platforms; and (iv) the cloud-systems techniques (microservices, lightweight isolation) needed to run untrusted third-party code. The review below is organised along those lines and closes with the research gap this project addresses. Bracketed numbers refer to the References section at the end of this chapter.

LLM-based autonomous agents

Two peer-reviewed surveys frame the field. Wang et al. [1] (Frontiers of Computer Science, 2024) propose a unified agent architecture of **profile, memory, planning and action** modules and catalogue applications in social science, natural science and engineering. Xi et al. [2] (Science China Information Sciences, 2025) describe agents as a "brain" (the LLM), "perception" and "action" components and discuss single-agent, multi-agent and human-agent settings. Both identify reliability, long-horizon planning and safe tool use as open problems - exactly the properties a solopreneur needs before delegating business operations. The ReAct paradigm [3] (ICLR 2023) interleaves reasoning traces with actions and is the control loop adopted by most production agent frameworks, including the LangGraph-based runtime planned for this project. Reflexion [10] (NeurIPS 2023) adds verbal self-reflection over failed attempts, improving task success without weight updates; we plan to use a similar critique step before an agent commits an irreversible business action.

Tool use and API integration

A marketplace agent is only useful if it can operate the merchant's existing SaaS stack. Toolformer [4] (NeurIPS 2023) showed that LLMs can learn when and how to call external APIs in a self-supervised way. ToolLLM [5] (ICLR 2024) scaled this to more than 16,000 real REST APIs and introduced a depth-first decision-tree search for multi-step tool chains, while Gorilla [6] (NeurIPS 2024) showed that retrieval of up-to-date API documentation sharply reduces hallucinated API calls. HuggingGPT [7] (NeurIPS 2023) is the closest academic analogue to our marketplace: a controller LLM selects and composes specialist models from a public hub based on their textual descriptions. Its results motivate our **machine-readable agent manifest** (capabilities, inputs, permission scopes), which lets both humans and other agents discover and compose marketplace agents. In industry these ideas have been standardised as the Model Context Protocol (MCP) for agent-to-tool connections [24] and the Agent2Agent (A2A) protocol for agent-to-agent delegation [25]; Hou et al. [17] survey the MCP ecosystem and its server life-cycle threats.

Multi-agent orchestration

AutoGen [8] (COLM 2024) provides conversable agents that cooperate through message passing and allow human participation at configurable points, and MetaGPT [9] (ICLR 2024) encodes standard operating procedures into role-specialised agents, reducing cascading hallucinations in multi-step work. Both demonstrate that decomposing work into narrow, specialised agents outperforms one general agent - the core premise of a marketplace of single-purpose agentic microservices. Our orchestrator adopts the explicit-workflow style of MetaGPT for deterministic business processes and AutoGen-style human-in-the-loop checkpoints for high-impact actions.

Evaluating agents on realistic tasks

AgentBench [11] (ICLR 2024) evaluated LLMs across eight interactive environments and found a large gap between commercial and open models, with long-term reasoning and instruction following as the main failure causes. τ -bench [12] (ICLR 2025) is directly relevant to our customer-support and order-management agents: it simulates a user, a domain policy and tool APIs (retail and airline) and introduces the pass@k metric, showing that even state-of-the-art function-calling agents succeed on fewer than half of tasks and are highly inconsistent across repeated trials. This motivates our **certification pipeline**: every published agent must pass a repeated-trial, policy-compliance test suite before listing, and its pass@k score is shown to buyers.

Security of agents and LLM app stores

Greshake et al. [13] (ACM AISec 2023) introduced indirect prompt injection, in which instructions hidden in data retrieved by an agent (an email, a web page, a support ticket) hijack its behaviour. AgentDojo [14] (NeurIPS 2024 Datasets & Benchmarks) offers 97 realistic tasks and 629 security test cases for measuring attacks and defences on tool-using agents. Deng et al. [15] (ACM Computing Surveys, 2025) organise agent threats into unpredictable user input, complex internal execution, variable operational environments and untrusted external entities. Most importantly for a marketplace, Hou, Zhao and Wang [16] (IEEE S&P 2025) analysed LLM app stores and found apps with misleading descriptions, privacy-policy violations, abusive content and malicious behaviour, concluding that stores need stronger vetting. The OWASP Top 10 for LLM Applications [31] adds practical controls such as least-privilege tool scopes and human approval for excessive agency. Our design responds with per-agent permission scopes, sandboxed execution, a policy engine with spending limits, an immutable audit log and pre-listing red-team tests.

Software marketplaces, platform economics and agentic markets

Rochet and Tirole [19] (Journal of the European Economic Association, 2003) model two-sided platforms in which pricing must attract both sides; our marketplace is two-sided (agent developers and solopreneurs), so revenue share and free tiers are design

decisions rather than afterthoughts. Martin et al. [20] (IEEE Transactions on Software Engineering, 2017) survey app-store analysis and show how ratings, reviews and release data feed back into software engineering - signals we expose to developers through usage analytics. Bansal et al. [18] (Microsoft Research, 2025) released Magentic Marketplace, a simulation of markets in which LLM agents act for buyers and sellers; they found that models exhibit a strong first-proposal bias (10-30x advantage for speed over quality) and are sensitive to the search mechanism. This informs our discovery design: ranking is based on certified quality metrics, not just recency. For the dynamic-pricing reference agent, den Boer [23] (Surveys in Operations Research and Management Science, 2015) reviews dynamic pricing with demand learning and provides the baseline (explore-exploit pricing with guardrails) against which an LLM-driven pricing agent can be compared.

Systems foundations

Dragoni et al. [21] describe microservices as small, independently deployable services communicating by messages, and note the matching trade-offs in operational complexity; our marketplace treats each agent as such a service. Firecracker [22] (USENIX NSDI 2020) shows that lightweight virtual machines can give strong isolation with start-up times around 125 ms and low memory overhead, which makes per-tenant isolation of untrusted agent code economically feasible. For the prototype we use gVisor-sandboxed containers on Kubernetes and keep Firecracker microVMs as the hardened option for 295B.

Research gap

Existing academic work studies agents, tool use, evaluation and security largely in isolation, and existing commercial marketplaces (see State of the Art) target enterprises or consumers. No published system combines (a) a developer-facing publishing pipeline with automated certification, (b) least-privilege, sandboxed execution against a merchant's own SaaS accounts, (c) human-in-the-loop approval tuned for a single operator, and (d) transparent reliability metrics (pass^k, cost per task) for buyers who cannot evaluate agents themselves. This project builds and evaluates such a system for the micro-SaaS solopreneur segment.

State-of-the-Art Summary

Industry has moved quickly from chat assistants to marketplaces of agents. OpenAI launched the **GPT Store** in January 2024 [28] for consumer-facing custom GPTs; Salesforce launched **AgentExchange** in March 2025 [27] as a marketplace of agents, actions and templates for its Agentforce platform; and AWS added an **AI Agents and Tools** category to AWS Marketplace in July 2025 [26], alongside AgentCore for running agents. On the protocol side, MCP [24] has become the common way to connect agents to tools, A2A [25] standardises agent discovery and delegation through "Agent Cards", and agent payment protocols - Stripe and OpenAI's Agentic Commerce Protocol (ACP), Google's Agent Payments Protocol (AP2) and Coinbase's x402 - are emerging for

agents that transact on a user's behalf [29]. Developer frameworks (LangGraph, AutoGen, CrewAI, OpenAI Agents SDK) make agents easy to build but leave distribution, trust and billing to the developer.

Table 1. Leading industry offerings compared with this project

Offering	Primary customer	What it provides	Gap for solopreneurs
OpenAI GPT Store [28]	Consumers, prosumers	Catalogue of custom GPTs inside ChatGPT	Conversational, not autonomous; limited back-office integration; weak vetting [16]
Salesforce AgentExchange [27]	Salesforce enterprise customers	Partner agents, actions, templates for Agentforce	Requires the Salesforce platform and enterprise budget
AWS Marketplace AI Agents & Tools [26]	Enterprise buyers on AWS	Procurement of agents, MCP servers and tools through AWS billing	Assumes cloud and DevOps expertise; procurement-oriented
MCP server registries [17][24]	Developers	Discovery of tool servers	Tools, not end-to-end agents; no certification, billing or monitoring
Agent frameworks (LangGraph, AutoGen [8], CrewAI)	Developers	Libraries to build agents	No distribution channel, trust layer or buyer-facing UI
This project	Micro-SaaS solopreneurs and indie developers	Certified agents, sandboxed execution against the merchant's SaaS, HITL approvals, usage billing and reliability metrics	-

Leading-edge techniques we will adopt: MCP for tool access, A2A Agent Cards for capability manifests, policy-guarded tool calling with human approval for high-impact actions, repeated-trial evaluation (pass^k) from τ -bench [12], prompt-injection testing in the style of AgentDojo [14], LLM observability (OpenTelemetry traces, Langfuse) and model routing across providers to control cost.

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Chapter 2. Project Justification

The problem. Solo founders are now a large part of the economy: the U.S. Census Bureau counts about 30 million nonemployer businesses, which generated roughly \$1.7 trillion in receipts in 2022 [30]. A micro-SaaS solopreneur must personally handle support tickets, pricing, invoicing, sales-tax and VAT obligations, reporting and marketing. These tasks are repetitive and rule-driven yet still need judgement, so they consume the founder's time without being easy to automate with traditional if-this-then-that tools.

Why now. LLM agents can now plan, call real APIs and recover from errors [1][3][5], and the industry has standardised how agents connect to tools (MCP) and to each other (A2A) [24][25]. Large vendors have responded with agent marketplaces, but these target enterprises already on Salesforce or AWS [26][27], or consumers chatting inside ChatGPT [28]. The long tail of independent businesses has no trustworthy place to obtain agents that act on their own Stripe, Shopify, Gmail or help-desk accounts.

Why a marketplace and not a single product. Each business function needs deep domain knowledge (for example, tax rules across jurisdictions). Research shows that narrow, specialised agents outperform general ones [8][9]. A marketplace lets many independent developers each build one excellent agent while the platform supplies the hard, shared parts: identity, credential custody, sandboxing, certification, approvals, billing and monitoring. This is the classic two-sided platform opportunity [19].

Why trust is the core technical contribution. Delegating real business actions to third-party agents is risky. Agents remain unreliable across repeated trials [12], are vulnerable to indirect prompt injection from the very emails and tickets they process [13][14], and LLM app stores already contain misleading and malicious apps [16]. The project's novelty is therefore not just a catalogue but a **trust layer**: (1) mandatory certification with repeated-trial reliability (pass^k) and prompt-injection tests before listing; (2) least-privilege permission scopes and sandboxed execution; (3) policy limits and human-in-the-loop approval for irreversible actions; and (4) transparent reliability and cost metrics shown to buyers, countering the first-proposal and speed biases observed in agentic markets [18].

Expected impact. For solopreneurs: lower operating cost and hours reclaimed from routine operations, with the owner kept in control. For developers: a distribution channel and revenue for specialised agents. For research: an open, measurable testbed that answers questions the literature leaves open - how certification metrics correlate with production reliability, how much human approval is needed for acceptable risk, and which defences stop prompt injection in multi-tenant agent platforms.

Publication and funding potential. We target a derivative paper on "Certifying third-party LLM agents for multi-tenant marketplaces" (candidate venues: an IEEE or ACM software-engineering or AI-systems workshop, or a conference such as IEEE SOSE/ICSOC), reporting pass^k reliability, attack-success rates before and after our

defences, and cost per task across model providers. The working prototype, three certified reference agents and a usability study with real solopreneurs make the project suitable for presentation to potential funders at the SJSU Expo.

Chapter 3. Project Requirements

Requirements

Actors and scope

Table 2. Actors

Actor	Description	Main goals
Solopreneur (Buyer)	Owner-operator of a micro-SaaS or small online business; non-expert in AI	Find trustworthy agents, connect them to business accounts, keep control, see results and cost
Agent Developer (Seller)	Independent developer who builds a specialised agent	Publish, version and price agents; get paid; see usage and failures
Platform Administrator	Project team operating the marketplace	Certify agents, handle abuse reports, monitor platform health
External SaaS Systems	Stripe, Shopify, Gmail, Zendesk, QuickBooks (sandbox accounts)	Accessed by agents through MCP connectors with scoped OAuth tokens
LLM Providers	Anthropic, OpenAI, open-weight models	Reached only through the platform LLM gateway

Use cases

Table 3. Use-case catalogue

ID	Use case	Primary actor	Summary
UC-01	Register and sign in	All	Create account, verify email, sign in with SSO; choose Buyer and/or Developer role
UC-02	Publish agent	Developer	Upload agent package and manifest (capabilities, required connectors, permission scopes, pricing); submit for certification
UC-03	Certify agent	Administrator / pipeline	Automated static scan, sandbox functional tests, pass^k reliability and prompt-injection tests; approve or reject
UC-04	Discover agents	Solopreneur	Search by natural language or filters; compare reliability, cost and permissions
UC-05	Install and configure agent	Solopreneur	Grant scoped OAuth access to business systems, set policies (spend limits, approval rules, schedule)
UC-06	Run agent	System / Solopreneur	Trigger on schedule, webhook event or on demand; agent executes in sandbox
UC-07	Approve high-impact action	Solopreneur	Review a proposed action (e.g., refund > \$100, price change > 10%) and approve, edit or reject
UC-08	Monitor activity	Solopreneur	View run history, actions taken, outcomes, LLM cost and errors
UC-09	Billing and payouts	Solopreneur / Developer	Subscribe or pay per use; developer receives revenue share
UC-10	Revoke / pause agent	Solopreneur	Immediately pause an agent and revoke its credentials
UC-11	Rate and report agent	Solopreneur	Leave rating or abuse report; triggers admin review
UC-12	Delegate to another agent	Agent	An installed agent discovers and invokes another certified agent through A2A, within the buyer's policies

Functional requirements (user stories)

Priorities follow the template: **E = Essential** (required for a working system at the end of 295B), **D = Desired** (planned and scheduled), **O = Optional** (stretch goals). Each requirement is uniquely identified, verifiable by the acceptance criterion given, and traceable to a use case, following the characteristics of a good SRS in IEEE 830 section 4.3.

Table 4. Functional requirements

ID	User story	Pri.	Trace	Acceptance criterion
FR-01	As a user, I want to register and sign in with email or Google SSO so that my account is secure.	E	UC-01	OIDC sign-in works; passwords never stored in plain text; sessions expire after 24 h
FR-02	As a user, I want role-based access (Buyer, Developer, Admin) so that each sees only permitted functions.	E	UC-01	Unauthorised API calls return HTTP 403; verified by automated tests for every endpoint
FR-03	As a developer, I want to scaffold an agent with the SDK/CLI so that I can start from a working template.	E	UC-02	agentmkt init creates a runnable agent and manifest; agentmkt run --local passes sample test
FR-04	As a developer, I want to publish an agent with a manifest declaring capabilities, connectors, scopes and price.	E	UC-02	Manifest validated against JSON Schema; invalid manifests rejected with field-level errors
FR-05	As a developer, I want semantic versioning of my agent so that buyers can stay on a stable version.	E	UC-02	New version does not affect existing installs until buyer upgrades; rollback to prior version possible
FR-06	As the platform, I want every submitted version certified before listing.	E	UC-03	Pipeline runs static scan, functional tests, pass ^k (k=4) and injection suite; result and report stored
FR-07	As a solopreneur, I want to search agents in natural language and filter by category, price and rating.	E	UC-04	Query returns ranked results in < 1 s for 1,000 agents; filters combinable
FR-08	As a solopreneur, I want an agent detail page showing permissions, reliability score, cost estimate and reviews.	E	UC-04	All manifest scopes listed in plain language; pass ^k and avg. cost/run shown
FR-09	As a solopreneur, I want to connect my business accounts via OAuth with only the scopes the agent needs.	E	UC-05	Tokens stored encrypted in vault; agent cannot call APIs outside granted scopes (tested)
FR-10	As a solopreneur, I want to set policies: spend caps, action limits, working hours and approval thresholds.	E	UC-05, UC-07	Policy violation blocks the action and notifies user; covered by unit tests
FR-11	As a solopreneur, I want agents to run on schedule, on webhook events or on demand.	E	UC-06	Cron, webhook and manual triggers each start a run; retried up to 3 times on transient failure
FR-12	As a solopreneur, I want to approve, edit or reject high-impact actions before they execute.	E	UC-07	Pending actions appear in console and by email; nothing executes without approval when rule applies
FR-13	As a solopreneur, I want a run history with every tool call, outcome and LLM cost.	E	UC-08	Each run shows timeline of steps; data retained 90 days; exportable as CSV
FR-14	As a solopreneur, I want to pause an agent and revoke its credentials instantly.	E	UC-10	Pause takes effect within 5 s; subsequent tool calls fail closed
FR-15	As the platform, I want an immutable audit log of every agent action and approval.	E	UC-06, UC-07	Append-only; records actor, agent version, inputs hash, outcome and timestamp
FR-16	As a team, we want three certified reference agents: customer support, dynamic pricing, invoicing/tax.	E	UC-06	Each passes certification and completes its scripted demo scenario end-to-end
FR-17	As a solopreneur, I want to subscribe or pay per use, and as a developer receive revenue share.	D	UC-09	Stripe test-mode subscriptions, metered usage and Connect payouts reconcile to the cent
FR-18	As a developer, I want usage and failure analytics for my agents.	D	UC-08	Dashboard shows installs, runs, success rate, top errors per version
FR-19	As a solopreneur, I want to rate and report agents.	D	UC-11	Ratings shown on detail page; report creates admin ticket
FR-20	As an agent, I want to delegate subtasks to another certified agent via A2A within the buyer's policies.	D	UC-12	Delegation requires buyer's prior consent; delegated calls appear in audit log
FR-21	As a solopreneur, I want a weekly digest of what agents did and money/time saved.	D	UC-08	Email digest generated every Monday with key metrics
FR-22	As a solopreneur, I want natural-language setup	O	UC-05	Generated policy shown for confirmation before

ID	User story	Pri.	Trace	Acceptance criterion
	("handle refunds under \$50") converted into policies.			saving
FR-23	As a developer, I want a local simulator of buyer accounts for testing.	O	UC-02	Mock MCP servers for Stripe/Shopify/Zendesk available in SDK
FR-24	As a solopreneur, I want agents that pay for services on my behalf (ACP/AP2).	O	UC-06	Prototype payment through a test-mode agent payment protocol
FR-25	As a user, I want a mobile-friendly approval experience with push notifications.	O	UC-07	Approvals usable on 375 px wide screen; push via web notifications

Reference agent requirements

Table 5. Reference agents (Essential)

Agent	Behaviour	Connectors	Human approval required when
Customer-support agent	Classifies and answers tickets using the merchant's FAQ and order data; issues refunds and resends licence keys	Zendesk/Gmail, Stripe	Refund > configured limit; low-confidence answer; angry/escalation intent
Dynamic-pricing agent	Proposes price changes from sales velocity, competitor prices and sentiment within merchant bounds	Stripe/Shopify, web search	Any change > 10% or outside min/max bounds
Invoicing & tax agent	Generates invoices, computes sales tax/VAT from tax-rate service, flags filing deadlines	Stripe, QuickBooks, tax-rate API	Any filing or payment action (agent prepares, never files autonomously)

Non-functional requirements

Table 6. Non-functional requirements

ID	Category	Requirement (measurable)	Pri.
NFR-01	Security	All agent code runs in a sandbox with no network egress except the LLM gateway and the connectors in its manifest	E
NFR-02	Security	Credentials encrypted at rest (AES-256) and in transit (TLS 1.2+); never exposed to agent code or logs	E
NFR-03	Security	Prompt-injection attack success rate on our test suite below 5% for certified reference agents	E
NFR-04	Reliability	Reference agents achieve pass ⁴ ≥ 0.80 on their certification scenarios	E
NFR-05	Performance	Marketplace API p95 latency < 300 ms (excluding agent runs); search < 1 s	E
NFR-06	Scalability	Support 100 concurrent agent runs and 1,000 listed agents on the staging cluster	D
NFR-07	Availability	Staging environment 99% monthly uptime during 295B evaluation period	D
NFR-08	Cost	Average LLM cost per reference-agent run reported; target < \$0.05 for support tickets	D
NFR-09	Usability	New solopreneur installs and configures an agent in < 10 minutes; SUS score ≥ 70 in user study	E
NFR-10	Accessibility	Web UIs meet WCAG 2.1 AA for colour contrast and keyboard navigation	D
NFR-11	Observability	100% of agent runs traced end-to-end (OpenTelemetry) with cost and token counts	E
NFR-12	Privacy	Tenant data isolated; buyer can delete account and all data within 24 h	E
NFR-13	Maintainability	Back-end unit-test coverage $\geq 70\%$; CI runs on every pull request	E
NFR-14	Portability	Full stack starts locally with one docker compose up command	E
NFR-15	Compliance	Agents provide assistance only; tax/legal outputs carry a disclaimer and require human filing	E

Constraints and assumptions

- External systems are used in sandbox/test mode only (Stripe test mode, Shopify development stores, trial help-desk accounts); no real customer money moves during the project.
- LLM usage is funded through provider credits or student programmes; the LLM gateway enforces a monthly budget per environment.
- The marketplace is a research prototype; it is not offered to the public and does not provide legal or tax advice.
- The team has three members; scope is sized to three reference agents plus the platform.

Chapter 4. Dependencies and Deliverables

Dependencies

Only showstopper dependencies - those that would halt the project if unavailable - are listed, each with a mitigation.

Table 7. Showstopper dependencies (VK = Vineet Kumar, NG = Nikhil Hiro Ghind, SP = Sakshat Nandkumar Patil)

ID	Dependency	Why it is a showstopper	Mitigation	Owner
D-01	LLM API access and credits (Anthropic, OpenAI)	Agents cannot reason or act without a capable model	Apply for academic/startup credits; LiteLLM gateway allows switching to open-weight models (Llama via Ollama) locally	NG
D-02	Cloud hosting (AWS EKS/RDS or GCP equivalent) and credits	No shared staging environment for integration, demo and load tests	AWS Educate / GitHub Student credits; full stack also runs locally with Docker Compose and kind	VK
D-03	Sandbox accounts and OAuth apps for Stripe, Shopify, Gmail, Zendesk	Reference agents and connectors cannot be developed or demonstrated	Register developer/test accounts in S1; mock MCP servers as fallback	SP
D-04	Tax-rate data source (e.g., Stripe Tax test mode or a public rate API)	Invoicing & tax agent cannot compute correct taxes	Use Stripe Tax in test mode; fallback to static rate tables for selected jurisdictions	NG
D-05	Open-source agent stack: LangGraph, MCP SDK, A2A SDK, Temporal	Runtime and orchestration are built on these	Pin versions; wrap behind our SDK interfaces so one can be replaced	NG / VK
D-06	Sandboxing technology (gVisor; Firecracker for hardening)	Untrusted agent code cannot be run safely without isolation	Start with gVisor on Kubernetes; restricted containers with network policies as minimum	VK
D-07	Advisor approval of scope and evaluation plan	Work cannot be accepted for credit without it	Review requirements and schedule with Prof. Ranjan in early October	All
D-08	Solopreneur participants for usability study and IRB guidance if required	NFR-09 and research evaluation cannot be validated	Recruit from SJSU entrepreneurship community and indie-maker forums from S8; confirm IRB needs early	SP

Deliverables

Table 8. Project deliverables

#	Deliverable	Description
1	Marketplace web application	Developer Portal and Solopreneur Console (Next.js), deployed to cloud staging
2	Marketplace back-end services	Registry, catalog/search, certification, billing, orchestration services with REST APIs (OpenAPI spec)
3	Agent runtime and policy engine	Sandboxed execution plane with LLM gateway, guardrails and human-in-the-loop approvals
4	Agent Developer SDK and CLI	Python package to scaffold, test locally, package and publish agents; published on GitHub with docs
5	Three certified reference agents	Customer support, dynamic pricing, invoicing & tax
6	MCP connector set	Stripe, Shopify, Gmail, Zendesk, QuickBooks connectors with least-privilege scopes
7	Certification and evaluation harness	Automated pipeline with pass^k reliability and prompt-injection test suites; reusable benchmark dataset
8	Monitoring dashboards	Run history, cost and reliability dashboards for buyers; platform health dashboards for admins

#	Deliverable	Description
9	Infrastructure as code	Terraform + Helm charts and GitHub Actions pipelines to recreate the environment
10	Documentation	Architecture and API docs, developer guide, user guide, threat model
11	Evaluation report and dataset	Reliability, security, performance, cost and usability results
12	Derivative research paper	Paper on certifying third-party LLM agents for multi-tenant marketplaces, submitted to a peer-reviewed workshop or conference (subject to advisor agreement)
13	Demo video and Expo prototype	5-minute end-to-end demo and live booth demo

Chapter 5. Project Architecture

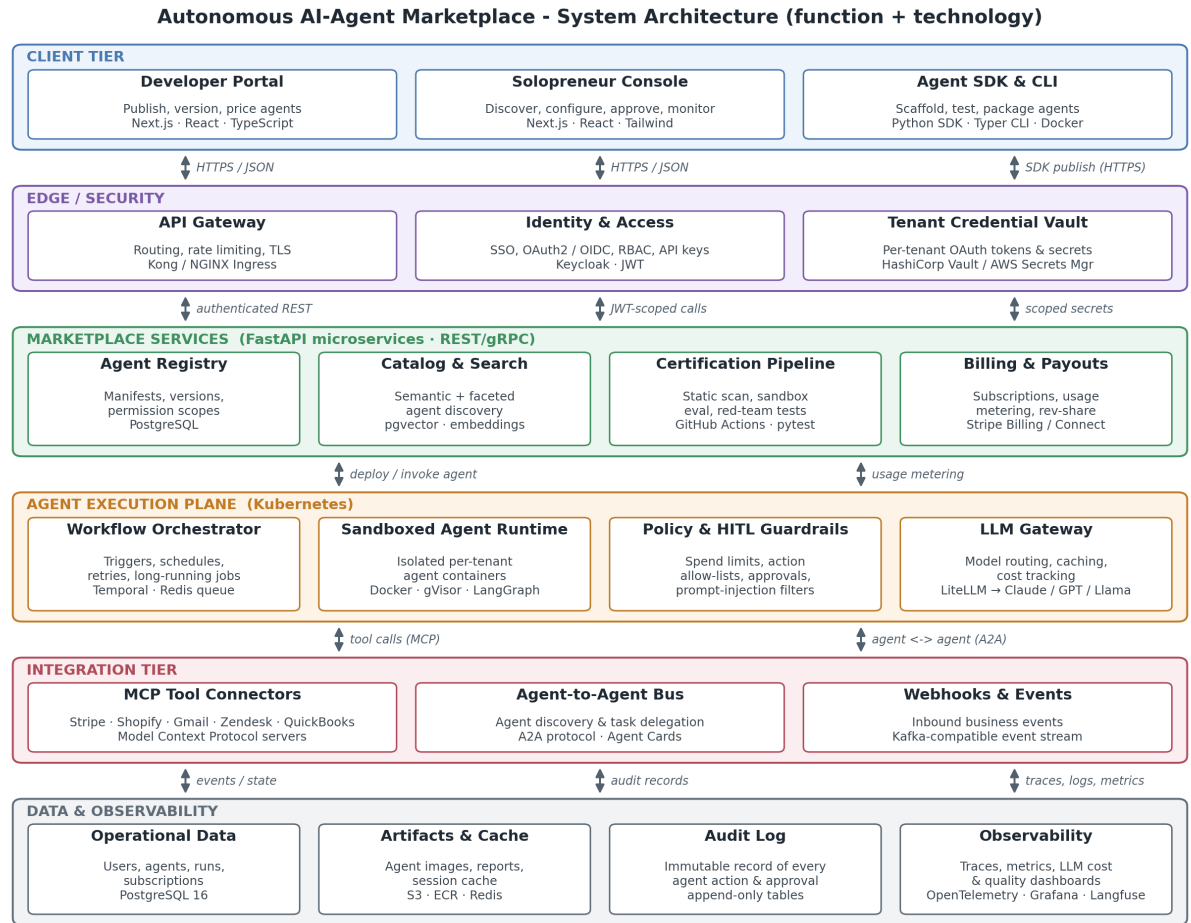


Figure 1. System architecture showing each component's function and technology

Architectural style. The system is a **layered, service-oriented (microservices) architecture with an event-driven execution plane**. Each tier has one responsibility and communicates with the next through authenticated REST/gRPC calls or events, so tiers can be developed by different team members and scaled independently [21].

Client tier. The *Developer Portal* lets developers register agents, upload versions, set prices and read certification reports and analytics. The *Solopreneur Console* is where buyers discover agents, connect accounts, set policies, approve pending actions and monitor runs. Both are Next.js/React/TypeScript apps. The *Agent SDK & CLI* (Python) scaffolds agents, runs them locally against mock connectors and publishes signed packages.

Edge and security tier. An *API gateway* (Kong or NGINX Ingress) terminates TLS, rate-limits and routes calls. *Identity & Access* (Keycloak, OAuth2/OIDC) issues JWTs carrying the user's role and tenant. The *Tenant Credential Vault* (HashiCorp Vault

or AWS Secrets Manager) holds each buyer's OAuth tokens; agents never see raw credentials - connectors receive short-lived, scope-restricted tokens at call time.

Marketplace services (Python FastAPI microservices over PostgreSQL). The *Agent Registry* stores manifests, versions and permission scopes. *Catalog & Search* combines keyword facets with semantic search over manifest embeddings (pgvector), ranking by certified reliability rather than recency to avoid the biases reported in [18]. The *Certification Pipeline* (GitHub Actions + pytest) runs static analysis, functional tests in the sandbox, repeated-trial pass^k tests [12] and prompt-injection tests [14]. *Billing & Payouts* uses Stripe Billing for subscriptions and metered usage and Stripe Connect for developer revenue share.

Agent execution plane (Kubernetes). The *Workflow Orchestrator* (Temporal) turns schedules, webhooks and manual triggers into durable runs with retries and timeouts. Each run executes in the *Sandboxed Agent Runtime*: a gVisor container per tenant and agent version, with network policies that allow egress only to the LLM gateway and approved connectors. Agents are built on LangGraph using a ReAct-style loop [3]. Every proposed tool call passes through the *Policy & HITL Guardrail* service, which checks scopes, spend limits and approval rules, filters suspected injected instructions, and pauses the workflow until the buyer approves when required. The *LLM Gateway* (LiteLLM) routes to Anthropic Claude, OpenAI GPT or open-weight models, caches prompts and records token cost per tenant.

Integration tier. *MCP tool connectors* wrap Stripe, Shopify, Gmail, Zendesk and QuickBooks as MCP servers [24] so any agent uses one uniform tool interface. The *Agent-to-Agent bus* implements A2A [25] so an agent can discover a certified agent through its Agent Card and delegate a task within the buyer's policies. *Webhooks & Events* ingest business events (new ticket, new order) into an event stream that triggers runs.

Data and observability tier. PostgreSQL holds users, agents, installations, runs and subscriptions; S3/ECR store agent images and generated artefacts; Redis caches sessions and queues. An append-only *audit log* records every action and approval. OpenTelemetry traces flow to Prometheus/Grafana for platform metrics and to Langfuse for LLM-level traces, cost and quality evaluation.

Key flow - running an agent. (1) A Zendesk webhook arrives at the gateway; (2) the orchestrator starts a run for the buyer's installed support agent; (3) the runtime loads the agent version in a sandbox and calls the LLM gateway; (4) the agent proposes a refund tool call; (5) the policy engine sees the amount exceeds the buyer's \$50 limit and pauses the run; (6) the buyer approves in the Console; (7) the Stripe connector executes with a short-lived scoped token; (8) the outcome, cost and trace are written to the audit log and dashboard, and usage is metered for billing.

Deployment. Local development uses Docker Compose and kind (Kubernetes in Docker); staging runs on AWS EKS with RDS PostgreSQL, provisioned with Terraform and Helm and deployed by GitHub Actions.

Chapter 6. Project Design

To be completed in Workbook Assignment 2 (UML class and sequence diagrams, UI mockups for the Developer Portal and Solopreneur Console, database ER diagram, agent manifest schema).

Chapter 7. QA, Performance, Deployment Plan

To be completed in Workbook Assignment 2 (test strategy, certification and red-team plan, performance and cost benchmarks, deployment pipeline).

Chapter 8. Implementation Plan and Progress

Implementation follows a **walking-skeleton** strategy: first make the thinnest end-to-end path work (publish an agent -> discover it -> install it -> run it in a sandbox -> see the result), then deepen each component in later sprints. Status values below reflect the state of the repository on the date of this workbook; the progress log is updated at the end of every sprint.

Programming and Execution Environment

Table 9. Programming and execution environment

Layer	Technology (version)	Purpose	Owner	Status
Source control & CI	GitHub monorepo, GitHub Actions	Code hosting, pull-request reviews, CI on every PR	VK	Repo created (Sep 1); CI in S1
Back-end services	Python 3.12, FastAPI, SQLAlchemy, Pydantic v2	Marketplace microservices and APIs	VK	S1
Agent runtime	LangGraph, MCP Python SDK, A2A SDK, LiteLLM	Agent loop, tool calls, delegation, model routing	NG	S1 spike
Front-end	Node 20, Next.js 15, React, TypeScript, Tailwind	Developer Portal and Solopreneur Console	SP	S1
Data	PostgreSQL 16 + pgvector, Redis 7	Operational data, semantic search, cache/queue	VK	S1
Orchestration	Temporal	Durable, retryable agent workflows	VK	S4
Identity & secrets	Keycloak, HashiCorp Vault (dev mode)	OIDC sign-in, RBAC, credential custody	SP	S2
Containers	Docker, Docker Compose, kind, gVisor	Local stack and sandboxed agent execution	VK / NG	S1-S3
Cloud	AWS EKS, RDS, S3, ECR; Terraform, Helm	Shared staging environment	VK	S5
Observability	OpenTelemetry, Prometheus, Grafana, Langfuse	Traces, metrics, LLM cost	SP	S5 / S7

Development Tools per Team Member

Table 10. Development tools acquired per member (target: all installed by end of Sprint 1, Oct 18)

Tool	Nikhil Hiro Ghind	Vineet Kumar	Sakshat N. Patil
IDE: VS Code / Cursor with Python, TS, Docker extensions	Yes	Yes	Yes
Docker Desktop, kind, kubectrl, Helm	Yes	Yes	Yes
Python 3.12 + uv, Poetry; pytest	Yes	Yes	Yes
Node 20 + pnpm; Playwright	-	-	Yes
LLM API keys (Anthropic, OpenAI) via shared team gateway	Yes	Yes	Yes
Ollama with an open-weight model for offline testing	Yes	-	-
AWS account (IAM user), Terraform CLI	-	Yes	Read-only
Stripe CLI + test account; Shopify Partner dev store	Yes	-	Yes
Postman / Bruno for API testing	Yes	Yes	Yes
Figma (UI mockups)	-	-	Yes

Tool	Nikhil Hiro Ghind	Vineet Kumar	Sakshat N. Patil
Project tracking: GitHub Projects board	Yes	Yes	Yes

Prototyping and Simulation Environment

The project is software-only, so the "hardware prototyping" environment is a reproducible simulated business:

- **Local cluster:** Docker Compose for services plus a kind cluster with gVisor runtime class to exercise sandboxing and network policies on a laptop.
- **Simulated solopreneur business ("DemoSaaS"):** a Stripe test-mode account with products, subscriptions and ~500 synthetic customers; a Shopify development store; a Zendesk trial seeded with ~300 synthetic support tickets generated from public FAQ-style templates.
- **Mock MCP servers** that replay recorded API responses, so tests are deterministic and cost-free.
- **Evaluation sandbox** adapting τ -bench-style retail scenarios [12] and AgentDojo-style injection cases [14] to our connectors.
- **Cloud staging** on AWS EKS for integration, load tests (k6/Locust) and the demo.

Sample Programs and Technology Spikes

Table 11. Sample programs used to learn each technology

Spike	Goal / what it proves	Owner	Target
SP-1 Hello agent	LangGraph ReAct agent calls a local MCP server (Stripe test mode: list customers) through LiteLLM	NG	Oct 18
SP-2 Sandboxed run	Run SP-1 inside a gVisor container on kind with egress limited by NetworkPolicy	NG / VK	Oct 18
SP-3 Registry API	FastAPI service stores and validates an agent manifest (JSON Schema) in PostgreSQL	VK	Oct 18
SP-4 Durable workflow	Temporal workflow that pauses for a human approval signal and resumes	VK	Oct 25
SP-5 Auth + console	Next.js app signs in via Keycloak OIDC and calls a protected API	SP	Oct 18
SP-6 OAuth connect	Console completes Stripe/Shopify OAuth and stores token in Vault	SP	Oct 25
SP-7 A2A call	One agent discovers another's Agent Card and delegates a task	NG	Nov 1

Draft agent manifest used by SP-3 (simplified):

```
name: support-refund-agent
version: 0.1.0
capabilities: [answer_ticket, issue_refund]
```

```

connectors:
  - zendesk: [tickets.read, tickets.write]
  - stripe: [charges.read, refunds.write]
policies_required: [refund_limit_usd, approval_channel]
pricing: {model: per_run, usd: 0.02}
runtime: {image: ghcr.io/agentmkt/support:0.1.0, timeout_s: 120}

```

Prototype Implementation Plan

Table 12. Prototype increments (end of CMPE 295A)

Step	Increment	Technologies exercised	Sprint
1	Walking skeleton: monorepo, Compose stack, CI, health-check endpoints, empty UIs	GitHub Actions, Docker, FastAPI, Next.js	S1
2	Publish path: SDK -> manifest validation -> registry -> portal listing	SDK/CLI, Pydantic, PostgreSQL	S2
3	Discover & install: semantic search, detail page, OAuth connect, policy form	pgvector, Keycloak, Vault	S3-S4
4	Run path: trigger -> Temporal workflow -> sandboxed agent -> MCP tools -> result	Temporal, gVisor, LangGraph, MCP, LiteLLM	S3-S4
5	Approval path: policy engine pauses run; buyer approves in console	Temporal signals, WebSockets	S4 (basic) / S6
6	Visibility: run timeline, audit log, cost per run	OpenTelemetry, Langfuse	S5
7	Deploy prototype to EKS; demo customer-support agent end-to-end (Milestone M2)	Terraform, Helm, EKS	S5

Progress Log

Table 13. Implementation progress (updated each sprint)

Date	Item	Status
Sep 1, 2026	Project repository created; README with concept, goals, team and roadmap	Done
Sep 2026	Project concept and abstract agreed with advisor	Done
Sep 2026	Literature search, state-of-the-art review, requirements, architecture (this workbook)	Done
Oct 5-18, 2026	Sprint 1: environment, tools, spikes SP-1, SP-2, SP-3, SP-5	Planned
Oct 19-Nov 1, 2026	Sprint 2: publish path; spikes SP-4, SP-6, SP-7	Planned

Chapter 9. Project Schedule

The project uses two-week Scrum sprints across CMPE 295A (Fall 2026) and CMPE 295B (Spring 2027), tracked on a GitHub Projects board. Each sprint ends with a demo to the team and a progress-log update. Every member owns one stream of work in every sprint, so tasks do not overlap; integration is done jointly at the end of each sprint.

Table 14. Milestones

Milestone	Date	Exit criteria
M1 Architecture baseline	Oct 4, 2026	Requirements, architecture and schedule reviewed with advisor
M2 End-to-end prototype	Dec 11, 2026	Customer-support agent published, discovered, installed and run in sandbox on cloud staging
M3 Beta	Mar 7, 2027	Three certified reference agents, approvals, billing and dashboards working
M4 Final release	May 7, 2027	Hardened, evaluated system; paper draft; demo video; Expo ready

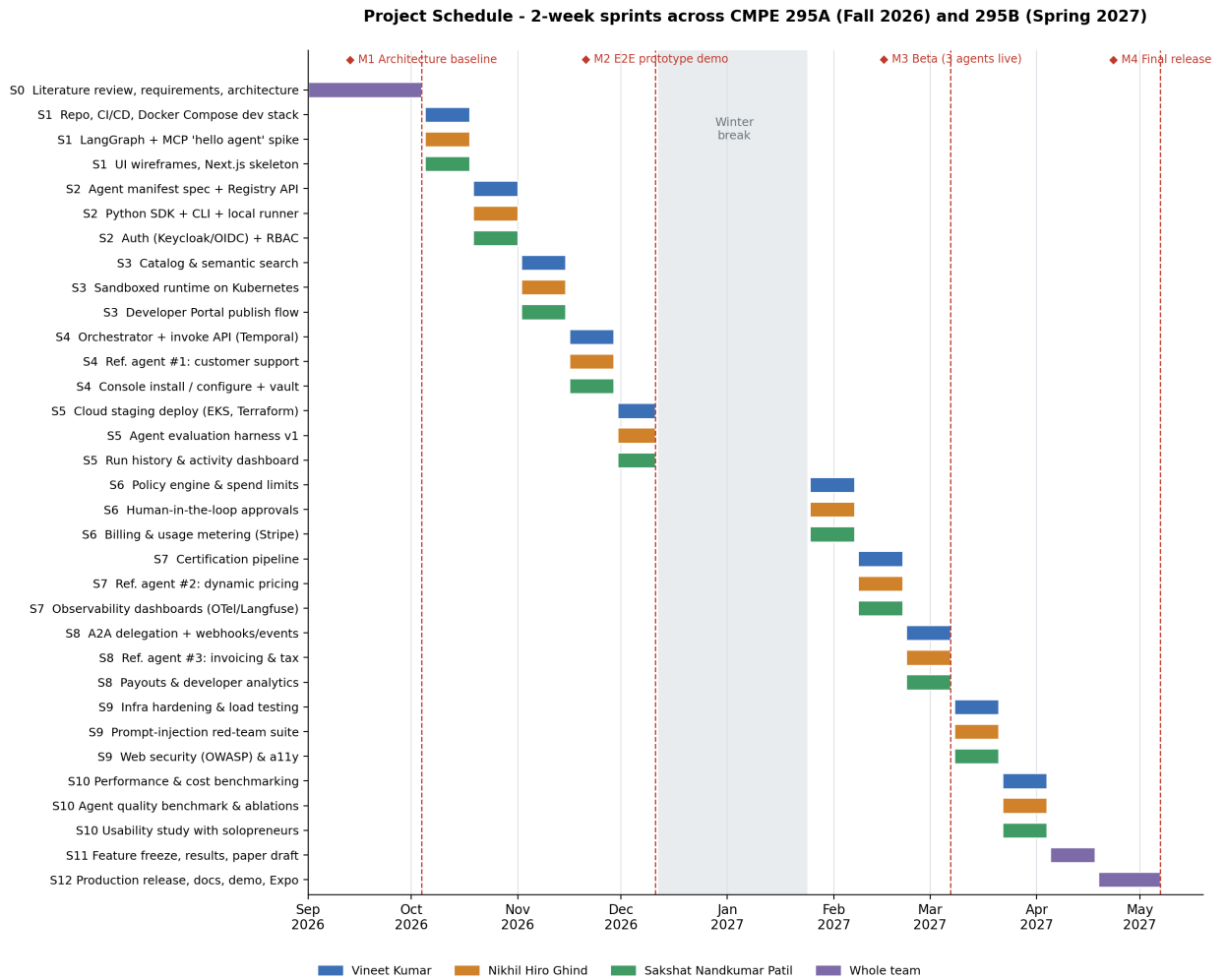


Figure 2. Project schedule (Gantt chart) by sprint and team member

Table 15. Task assignment per sprint (no overlapping tasks between members)

Sprint (dates)	Vineet Kumar	Nikhil Hiro Ghind	Sakshat N. Patil
S0 (Sep 1-Oct 4)	Systems & platform literature; architecture	Agent literature; agent requirements	Security literature; state of the art; requirements
S1 (Oct 5-18)	Monorepo, CI/CD, Compose stack, Postgres	LangGraph + MCP hello agent; gVisor spike	Wireframes; Next.js skeleton; Keycloak spike
S2 (Oct 19-Nov 1)	Manifest schema; Registry API	Python SDK, CLI, local runner	Auth, RBAC, OAuth connect, Vault
S3 (Nov 2-15)	Catalog & semantic search	Sandboxed runtime on Kubernetes	Developer Portal publish flow
S4 (Nov 16-29)	Temporal orchestrator; invoke API	Reference agent #1: customer support	Console install/configure; policy form
S5 (Nov 30-Dec 11)	EKS staging (Terraform/Helm)	Evaluation harness v1 (pass^k)	Run history & activity dashboard
Winter break	Design docs for Workbook 2 (optional)	Threat model (optional)	UI mockups v2 (optional)
S6 (Jan 25-Feb 7)	Policy engine & spend limits	Human-in-the-loop approvals	Billing & usage metering (Stripe)
S7 (Feb 8-21)	Certification pipeline	Reference agent #2: dynamic pricing	Observability dashboards
S8 (Feb 22-Mar 7)	A2A delegation; webhooks/events	Reference agent #3: invoicing & tax	Payouts; developer analytics
S9 (Mar 8-21)	Infrastructure hardening; load tests	Prompt-injection red-team suite	Web security (OWASP) &

Sprint (dates)	Vineet Kumar	Nikhil Hiro Ghind	Sakshat N. Patil
			accessibility
S10 (Mar 22-Apr 4)	Performance & cost benchmarks	Agent quality benchmark & ablations	Usability study with solopreneurs
S11 (Apr 5-18)	Results: performance section of paper	Results: reliability/security sections	Results: usability section; figures
S12 (Apr 19-May 7)	Production release & runbooks	SDK docs & sample agents	User guide, demo video, Expo booth